

Diverse taxa of cyanobacteria produce β -N-methylamino-L-alanine, a neurotoxic amino acid



Cyanobacteria can generate molecules hazardous to human health, but production of the known cyanotoxins is taxonomically sporadic. For example, members of a few genera produce hepatotoxic microcystins, whereas production of hepatotoxic nodularins appears to be limited to a single genus. Production of known neurotoxins has also been considered phylogenetically unpredictable. We report here that a single neurotoxin, β -N-methylamino-L-alanine, may be produced by all known groups of cyanobacteria, including cyanobacterial symbionts and free-living cyanobacteria. The ubiquity of cyanobacteria in terrestrial, as well as freshwater, brackish, and marine environments, suggests a potential for wide-spread human exposure.